

Event-Nodes and Texture-Nodes: Extending Causal Mechanistic Interpretability with Irreversible Structure

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Abstract

Mechanistic interpretability has increasingly converged on a causal paradigm in which neural networks are analyzed through interventions, counterfactuals, and causal abstractions. While this framework successfully distinguishes correlation from mechanism, it remains implicitly state-centered and treats irreversibility as a contingent byproduct of information loss. This essay proposes a minimal, formal extension of the causalmechanistic framework by introducing two typed classes of causal nodes: *texture-nodes*, which carry instantaneous values, and *event-nodes*, which record regime selections that irreversibly constrain future admissible behavior. Drawing an explicit analogy to procedural material systemswhere generator selection and parameter ratios play distinct roleswe develop typing rules, a generalized causal tracing method, and graph rewrite semantics that preserve causal rigor while accounting for history sensitivity, structural irreversibility, and algorithmic regime switching. The result is a unified account in which causal abstraction remains valid, but is explicitly temporal, plural, and constraint-aware.

1 Introduction

Recent work in mechanistic interpretability has emphasized the necessity of causal reasoning for understanding neural networks. Interventions, rather than mere inspection, are taken as the criterion for distinguishing true mechanisms from correlational artifacts. Within this paradigm, understanding a model means reverse-engineering the algorithm it implements by demonstrating systematic interventional correspondence between low-level components and high-level computational structure.

Despite its success, this framework implicitly privileges instantaneous internal states and treats history as relevant only insofar as it manifests in present activations. Irreversibility appears primarily as epistemic degradation caused by noise or ablation, rather than as a constitutive feature of mechanism itself. As a result, several phenomena—commitment, collapse, regime switching, and substitution failure—remain awkward to express within a purely state-based causal graph.

This essay introduces a minimal extension that resolves these limitations without abandoning causal rigor. The core move is a typed distinction between two kinds of causal nodes: *texture-nodes*, which carry values, and *event-nodes*, which select generative regimes and irreversibly constrain

future behavior. The extension is minimal in the sense that it adds no new metaphysics, only new types for causal variables and interventions.

2 From State-Based Causality to Typed Causal Nodes

Standard causal models represent systems as directed acyclic graphs whose nodes carry variables updated by structural equations. Interventions replace the value of a variable, and causal responsibility is assessed by downstream changes in behavior.

We retain this structure but introduce a typing function that distinguishes value-carrying nodes from constraint-carrying nodes.

Definition 1 (Typed Causal Graph). *A typed causal model is a tuple*

$$\mathcal{M} = (G, \tau, \text{Val}, \text{Reg}, \text{Adm})$$

where $G = (V, E)$ is a directed graph, $\tau : V \rightarrow \{\mathbf{T}, \mathbf{E}\}$ assigns each node a type, Val is the space of texture values, Reg is the space of event regimes, and Adm maps realized histories to sets of admissible futures.

Texture-nodes ($\mathbf{T}$) represent internal quantities whose causal role is exhausted by their instantaneous value. Event-nodes ($\mathbf{E}$) represent structural choices whose effects are not reducible to a value, but instead alter which future states remain reachable.

This distinction parallels the difference between modifying parameters of a procedural generator and switching the generator itself.

3 Texture-Nodes

A texture-node carries a value $X_v \in \text{Val}_v$ and updates according to a structural equation

$$X_v := f_v(X_{\text{Pa}(v)}, E_{\text{Pa}(v)}).$$

Interventions on texture-nodes are standard do-operations:

$$\text{do}(v := x), \quad x \in \text{Val}_v.$$

Texture-nodes are fully replayable: their causal role can be tested by substituting values and observing behavioral changes. Steering vectors, activation patching, and difference-in-mean methods all operate at this level.

Importantly, texture values often arise from generator functions parameterized by ratio-like quantities. These ratios behave as vector directions in parameter space, but their meaning depends on the generator selected upstream.

4 Event-Nodes

An event-node carries a regime value $E_v \in \text{Reg}_v$ that selects a generative mode or structural configuration. Its update rule may depend on both texture and event parents:

$$E_v := g_v(X_{\text{Pa}(v)}, E_{\text{Pa}(v)}).$$

Interventions on event-nodes take the form

$$\text{do}(v := r), \quad r \in \text{Reg}_v.$$

Unlike texture interventions, event interventions need not be reversible. They can permanently restrict the space of admissible futures.

Definition 2 (Structural Irreversibility). *An intervention on an event-node e is structurally irreversible at time t if*

$$\text{Adm}(\mathcal{H}_t) \neq \text{Adm}(\mathcal{H}_t[R \leftarrow r']),$$

where $\mathcal{H}_t$ is the realized history up to t .

Irreversibility here is not informational but structural: even perfect replay cannot restore the original counterfactual space.

5 Generator Selection and Parameter Ratios

Consider a texture-node generated by a procedural function G_r selected by an event-node with regime r :

$$X = G_r(\rho(\theta), \dots),$$

where $\rho(\theta)$ denotes ratios of generator parameters.

Changing $\rho(\theta)$ modifies the realized texture without altering the generator family. Changing r selects a different generator altogether, producing a qualitatively different space of possible outputs. Parameter ratios thus behave as vectors only relative to a fixed regime.

This clarifies why some activation directions generalize while others fail under context shifts: the underlying generator has changed.

6 Causal Tracing with Event-Nodes

Standard causal tracing corrupts an input and restores internal activations to identify mediators. With typed nodes, two distinct restoration operations are available.

- **Texture restoration:** $\text{do}(X_v := X_v^*)$
- **Event restoration:** $\text{do}(E_v := E_v^*)$

Texture restoration measures recovery of output behavior. Event restoration measures recovery of admissible futures. A node is a structural mediator if restoring its regime reinstates counterfactual possibilities even when output accuracy remains unstable.

This explains why some interventions preserve fluency while destroying coherence: the texture is intact, but the regime has shifted.

7 Graph Rewrite Semantics

Event-nodes can be given explicit operational meaning as graph rewrite triggers. The texture computation graph contains labeled rewrite sites. A rewrite rule has the form

$$(s, r) \Rightarrow G_s^{(r)},$$

indicating that site s is replaced by subgraph $G_s^{(r)}$ when regime r is active.

Rewrites must preserve interface types, ensuring that regime switching is structural rather than destructive. Texture parameters live inside the selected subgraph; event-nodes determine which subgraph exists.

This formalizes regime switching as algorithmic structure, not noise.

8 Causal Abstraction Revisited

Causal abstraction is traditionally validated by interventional equivalence. Under the typed framework, abstraction requires two correspondences:

1. Texture correspondence: value-level interventions align.
2. Event correspondence: regime transitions and irreversibility align.

An abstraction that preserves outputs but misrepresents regime structure is behaviorally accurate but temporally incoherent. Multiple abstractions may remain valid, but each carries an explicit scope and loss profile.

9 What Counts as Understanding

Understanding is no longer mere control or prediction. It is the ability to reconstruct which regimes were selected, which futures were foreclosed, and how value-level variation unfolds within those constraints.

A system may pass all texture-level tests while remaining opaque at the event level. Such cases mark the boundary of state-based explanation, not its failure.

10 Conclusion

By distinguishing texture-nodes from event-nodes, the causalmechanistic paradigm is extended just enough to account for history sensitivity, irreversibility, and regime selection without abandoning its core commitments. Interventions remain the criterion of explanation, but their targets are now typed. Causal tracing becomes structural as well as behavioral. Graph rewrite rules make explicit what was previously implicit in algorithmic mixture.

The resulting framework preserves the strengths of causal interpretability while making explicit the limits of state-only analysis. Mechanisms are not merely paths through variables, but trajectories through constrained possibility spaces.